

Soft Spaces - Phase 3.2

Interim Technical Summary Report

Interim report - 10 September 2026

Phase 3.2 focus: from observed $\pm E$ organization to an explicit algebraic symmetry mechanism

Executive summary

Phase 3.2 has shifted the project from detecting a strong REAL-specific $\pm E$ organization to identifying an explicit algebraic mechanism behind that organization. The work to date establishes that the $\pm E$ symmetry is not a hotspot-local phenomenon: it is a global property of the frozen REAL model for the reference seed. A concrete Pauli-string symmetry generator Γ has been identified, and the full operator chain from Hamiltonian anticommutation to the second-order effective operator has been verified numerically to approximately 10^{-12} precision.

Current central result

For the frozen 8-qubit, 11-term model at seed 25042000, $\Gamma = \text{XIXXIIXX}$ satisfies $\Gamma H \Gamma^\dagger = -H$ exactly in floating-point arithmetic. The same Γ gives $\Gamma V_{\text{deph}} \Gamma^\dagger = -V_{\text{deph}}$ and $\Gamma V_{\text{trans}} \Gamma^\dagger = +V_{\text{trans}}$. This implies and numerically verifies $P_{-E} = \Gamma P_E \Gamma^\dagger$, $Q_{-E} = \Gamma Q_E \Gamma^\dagger$, $(PVQ)_{-E} = s \Gamma (PVQ)_E \Gamma^\dagger$, and $\text{Sigma}_{-E} = -\Gamma \text{Sigma}_E \Gamma^\dagger$.

1. Starting point of Phase 3.2

Phase 3.1 ended with a strong and reproducible $\pm E$ organization in REAL relative to NULL. The open question was no longer whether a structure existed, but what mathematical mechanism produced it. Phase 3.2 therefore adopted a mechanism-first hypothesis: the organization might arise from structured coupling between a complete degeneracy-closed P subspace and its complement Q.

$$\mathbf{P} \mathbf{V} \mathbf{Q}$$

Coupling from the selected eigenspace P into its complement Q.

$$\text{Sigma}_E = \mathbf{P} \mathbf{V} \mathbf{Q} \, g_{\text{eta}}(E - Q H Q) \, \mathbf{Q} \mathbf{V} \mathbf{P}$$

Regularized second-order effective operator; $g_{\text{eta}}(x) = x / (x^2 + \text{eta}^2)$.

2. v32_0 - first P/Q mechanism probe

The first diagnostic program examined the five strongest existing $\pm E$ hotspot pairs for seed 25042000. It measured $||\text{PVP}||_{\text{F}}$, $||\text{PVQ}||_{\text{F}}$, $||\text{PVQ}||_{\text{F}}^2$, the P-to-Q spectral gap, and $||\text{Sigma}_E||_{\text{F}}$, separately for REAL and NULL and for dephasing and transverse perturbations.

- REAL showed essentially exact $\pm E$ equality in $||\text{PVQ}||_{\text{F}}^2$ and $||\text{Sigma}_E||_{\text{F}}$, typically at 10^{-14} to 10^{-12} asymmetry.
- NULL showed large $\pm E$ asymmetry, often $O(10^{-1})$ in $||\text{PVQ}||_{\text{F}}^2$ and $O(1-10)$ or larger in $||\text{Sigma}_E||_{\text{F}}$.
- For seed 25042000, $||\text{PVP}||_{\text{F}}$ was near numerical zero in REAL for the selected hotspots; later data showed this was seed-dependent and not a universal explanation.
- A diagnostic bug was identified: the NULL gap initially reused the REAL eigenvalue complement. Sigma itself was already computed correctly from the diagonalized NULL QHQ operator.

3. v32_0_1 - global $\pm E$ symmetry control

The corrected control program fixed the NULL gap definition and expanded the analysis from five hotspot pairs to all 32 exact $\pm E$ pairs for seed 25042000. The pairs were retained in frozen Phase-3.1 prominence order and grouped into TOP, MIDDLE, and BOTTOM thirds. This directly tested whether the symmetry was hotspot-specific or global.

Quantity	REAL - all 32 pairs	NULL - all 32 pairs
Dephasing $ \Delta\text{Sigma} $	median 5.91e-14; max 1.12e-12	median 3.08; max 12.19
Transverse $ \Delta\text{Sigma} $	median 3.49e-13; max 2.49e-12	median 9.47; max 25.16
Dephasing $ \Delta\text{PVQ}^2 $	median 1.24e-14; max 1.10e-13	median 1.40e-1; max 3.37e-1
Transverse $ \Delta\text{PVQ}^2 $	median 9.77e-15; max 5.51e-14	median 1.70e-1; max 7.02e-1

Interpretation of v32_0_1

The $\pm E$ equality is global across the complete REAL pair set for the reference seed. Hotspots do not create the symmetry; they sit on top of a broader algebraic $\pm E$ structure.

The corrected NULL gaps also became clearly different from the REAL gaps, confirming that the earlier identical gap values were an implementation artifact rather than a physical result.

4. v32_1 - search for the symmetry generator

The next step searched the Pauli-string space for operators Γ that anticommute term-by-term with the frozen Hamiltonian. Thirty-two such candidates were found. This is already a structural result: the exact $\pm E$ spectral pairing has an explicit algebraic origin rather than being a numerical coincidence.

$$\Gamma H \Gamma^\dagger = -H$$

The candidate with the smallest projector-map residual was YIZXXIY, with zero Hamiltonian anticommutation residual and projector mapping residuals of order 10^{-12} . However, candidate XIXXIIXX was more useful for the perturbation analysis because it gave clean, exact transformation rules for both frozen perturbation families:

$$\Gamma = \text{XIXXIIXX}$$

$$\Gamma V_{\text{deph}} \Gamma^\dagger = -V_{\text{deph}}$$

$$\Gamma V_{\text{trans}} \Gamma^\dagger = +V_{\text{trans}}$$

5. v32_2 - operator derivation test

v32_2 stopped searching and instead tested the raw operator identities generated by $\Gamma = \text{XIXXIIXX}$. Gauge freedom in individual eigenvectors was avoided by using spectral projectors and full embedded operators. A fixed numerical tolerance was used only as a validation gate; the physically relevant evidence is the residual scale and the analytically predicted sign structure.

$$P_{-E} = \Gamma P_E \Gamma^\dagger$$

$$Q_{-E} = \Gamma Q_E \Gamma^\dagger$$

$$(P V Q)_{-E} = s \Gamma (P V Q)_E \Gamma^\dagger$$

5.1 Why Sigma has a fixed minus sign

The sign of the second-order effective operator is not ambiguous. Under Γ , both the Hamiltonian and the reference energy change sign. Therefore the energy denominator transforms with an overall minus sign. Because the regularization $g_\eta(x) = x/(x^2 + \eta^2)$ is an odd function, the regularized resolvent also changes sign. The two V factors contribute $s^2 = 1$, independently of whether the perturbation family is even or odd under Γ .

$$\Gamma (E - Q H Q) \Gamma^\dagger = - [-E - Q_- H Q_-]$$

$$\text{Sigma_}E = -\text{Gamma Sigma_}E \text{Gamma}^\dagger$$

5.2 Numerical closure

Identity / diagnostic	Result
Gamma H Gamma † = -H	relative residual 0.000000e+00
dephasing transformation	s = -1; residual 0.000000e+00
transverse transformation	s = +1; residual 0.000000e+00
P-map over 32 pairs	median 2.91e-13; max 1.60e-12
Q-map over 32 pairs	median 3.67e-14; max 2.02e-13
PVQ-map, dephasing	median 2.89e-13; max 1.60e-12
PVQ-map, transverse	median 2.91e-13; max 1.60e-12
Sigma minus-sign, dephasing	median 3.81e-13; max 1.60e-12
Sigma minus-sign, transverse	median 3.92e-13; max 1.61e-12
Wrong Sigma plus-sign	residual 2.0 for both families
All np.allclose assertions	passed

Closed result

For the reference seed, the observed equality of Frobenius norms at $\pm E$ is a consequence of the algebraic symmetry. It is not merely an empirical correlation.

6. What has and has not been explained

Phase 3.2 has now explained the global $\pm E$ organization, but it has not yet explained hotspot prominence. This distinction is essential.

Question	Status
Why does the REAL spectrum occur in exact $\pm E$ pairs?	Explained by $\Gamma H \Gamma^\dagger = -H$.
Why are PVQ norms paired at $\pm E$?	Explained by projector covariance and $\Gamma V \Gamma^\dagger = sV$.
Why are second-order Sigma norms paired at $\pm E$?	Explained by $\text{Sigma_}E = -\Gamma \text{Sigma_}E \Gamma^\dagger$.
Is $\text{PVP} \approx 0$ the general mechanism?	No. It occurs for seed 25042000 but not generally.
Why are some $\pm E$ pairs strong hotspots and others weak?	Open. This is now the main local-mechanism question.
Does the same Γ work unchanged for every frozen seed?	Not yet established in this interim report.

7. The emerging general formula structure

The work is moving toward a more general and less ad hoc formulation. The likely final description has two layers: a global symmetry law and a local hotspot-selection condition.

$$\text{Global layer: } \text{Gamma H Gamma}^\dagger = -H, \quad \text{Gamma V Gamma}^\dagger = s V$$

$$\text{Derived layer: } P_E = \text{Gamma P_}E \text{Gamma}^\dagger, \quad \text{Sigma_}E = -\text{Gamma Sigma_}E \text{Gamma}^\dagger$$

$$\text{Local layer: } \text{hotspot prominence} = F(\text{local P-Q coupling, gaps, cancellation, geometry, ...})$$

The global layer is now strongly constrained and numerically verified for the reference model. The local function F remains to be identified. This is a cleaner theoretical position than trying to force all behavior into one empirical hotspot formula.

8. Recommended next step

The next Phase 3.2 task should no longer be another $\pm E$ -symmetry proof. It should search for the local scalar or operator feature that varies along the already-symmetric $\pm E$ background and predicts the frozen prominence ranking. Candidate diagnostics should be evaluated against TOP/MIDDLE/BOTTOM prominence while preserving the established Γ symmetry.

Working question for the next session

Given that $\pm E$ covariance is global, what local property of the P-Q effective dynamics determines whether a pair becomes a strong hotspot?

9. Reproducibility record

- Frozen model: 8 qubits, 11-term Hamiltonian, Phase-3.1 namespace v30.2.
- Reference Hamiltonian seed used for the closed derivation chain: 25042000.
- Degeneracy tolerance: $1e-10$; exact $\pm E$ pairing tolerance: $1e-8$.
- Regularization: $\eta = 1e-3$ with $g_\eta(x) = x/(x^2 + \eta^2)$.
- Perturbation families: dephasing Z/ZZ and transverse X/XX, five terms per family.
- Programs completed: v32_0_mechanism_probe.py; v32_0_1_mechanism_probe.py; v32_1_symmetry_operator.py; v32_2_symmetry_derivation_test.py.

10. Interim conclusion

Phase 3.2 has moved from phenomenology to mechanism. The strong $\pm E$ organization first seen statistically is now tied to an explicit anticommuting Pauli symmetry. The operator identities are verified across all 32 exact $\pm E$ pairs for the reference seed, and the second-order effective operator carries the analytically required minus sign. The remaining scientific problem is narrower and more informative: to determine which local feature selects hotspot strength within this globally symmetric algebraic framework.